STROMA WORLD SPEC v1.0

Simulating the PDAC Stromal Barrier for Evolutionary Counterfactual Drug Discovery

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Field	Value
Version	1.0
Date	February 26, 2026
Author	Muniir (University of Louisville, Bioengineering PhD Program)
Platform	PhysiCell (C++) + Evolutionary Algorithm wrapper (Python)
HPC Target	University of Louisville LARCC
Cycle	1 of 3+ (Stroma Barrier Only)
Status	Phase 1 Complete — Specification Locked

> **One-sentence purpose:** Build a minimal simulation world where an evolutionary algorithm can discover interventions that breach the PDAC stromal barrier and kill the tumor cells behind it.
> **Scope boundary:** Stroma barrier formation and drug exclusion ONLY. No immune cells, no vasculature dynamics, no metabolic reprogramming, no metastasis. Those are Cycles 2–3.
> **Design philosophy:** If it doesn't contribute to barrier formation, drug exclusion, or treatment resistance, it's not in v1.0.

PHASE 1 DELIVERABLE INDEX

This integrated spec is accompanied by standalone deliverables. Together they constitute the complete Phase 1 package.

Standalone Deliverables (separate documents)

#	Deliverable	File	Status
D1	Gene Set (21-gene panel)	Stroma_World_Gene_Set_v1.0.md	✅
D2	Gene State Representation & Combination Rules	Stroma_World_Gene_State_Representation_v1.0.md	✅
D3	**Mutation→Behavior Class Map**	Stroma_World_Mutation_Behavior_Map_v1.0.md	✅
D4	Tumor Baseline Traits	Stroma_World_Tumor_Baseline_Traits_v1.0.md	✅
D5	**Tumor Calibration Knobs + Ranges**	Stroma_World_Tumor_Calibration_Knobs_v1.0.md	✅
D6	**Reality Anchors (10)**	Stroma_World_Reality_Anchors_v1.0.md	✅
D7	**Anchor Metrics + Pass/Fail**	Stroma_World_Anchor_Metrics_v1.0.md	✅
D8	Biological Modules (8)	Stroma_World_Biological_Modules_v1.0.md	✅
D9	**Module Wiring Edge List**	Stroma_World_Module_Edge_List_v1.0.md	✅
D10	Pathogenicity Scoring Guardrails	Stroma_World_Pathogenicity_Scoring_Guardrails_v1.0.md	✅

Integrated Spec Sections (this document)

#	Section	Content	Status
§1	Actors	3 actor types, goals, behaviors, state variables	✅
§2	World State Variables	Cell-level, environment, host-level variable registry	✅
§3	Rules of the World	69 numbered rules, 55 active, 14 deferred	✅
§4	Reality Check Suite	5 integration tests gating EA deployment	✅
§5	Assumptions & Exclusions	9 assumptions, 8 exclusions, risk-ranked	✅
§6	Version Label & Change Control	Version history and amendment protocol	✅

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§1: ACTORS

> Three actors. One builds the prison (tumor). One is the prison (stroma). One is the outside world (boundary). Everyone else arrives in later cycles.

Actor 1: PDAC Tumor Cell

Identity

The protagonist and the antagonist. This cell builds the barrier that protects it, creates the hypoxia that hardens it, and secretes the signals that recruit its own shield.

Goal

Proliferate. Survive. Secrete. In that priority order. The tumor cell has no explicit goal function — its behaviors are hardwired consequences of its genotype (KRAS=GOF, TP53=LOF, CDKN2A=LOF, SMAD4=LOF, BCL-XL=GOF).

Core Behaviors

Behavior	Module	Description
Constitutive paracrine secretion	M2	Secretes TGF-β and SHH at rates set by Knobs 1a, 1b. Always on. Amplified under hypoxia.
Rapid proliferation	M4	Divides fast with checkpoints broken. Paused only by mechanical compression or acute hypoxia.
Apoptosis resistance	M4	Low basal death rate. Dies only when intracellular drug is high AND efflux hasn't neutralized it.
Inducible EMT	M5	Transitions to mesenchymal phenotype under TGF-β or hypoxia. Reversible. Changes motility, adhesion, MMP2, PD-L1.
Hypoxia adaptation	M1	HIF1A activates below O_2 threshold → amplifies secretion, primes stress response.
Drug-inducible efflux	M7	ABCB1 silent at baseline. Activates via NRF2 after drug exposure + induction delay.
O_2 and drug consumption	Passive	Consumes O_2, uptakes drug when exposed.

State Variables

Variable	Type	Initial	Mutable?
cell_id	Integer	Unique	No
cell_type	Enum {TUMOR}	TUMOR	No

Variable	Type	Initial	Mutable?
position	(x,y) or (x,y,z)	Seeded	Yes
gene_state[21]	Array of {WT, GOF, LOF}	Canonical PDAC	No
hif1a_active	Boolean	OFF	Yes
zeb1_active	Boolean	OFF	Yes
cdh1_expressed	Boolean	ON	Yes
nrf2_active	Boolean	OFF	Yes
abcb1_active	Boolean	OFF	Yes
mmp2_active	Boolean	OFF	Yes
intracellular_drug	Continuous (0–1)	0	Yes
time_since_drug_exposure	Continuous (hours)	NULL	Yes
emt_extent	Categorical {NONE, LOW, MED, HIGH}	NONE	Yes
mechanical_pressure	Continuous	0	Yes

Actor 2: Pancreatic Stellate Cell / Cancer-Associated Fibroblast

Identity

One cell type, two modes. Quiescent PSC is the resting state. Activated CAF is the barrier-building state. The transition is the central event of stroma formation.

Goal

As PSC: Nothing. Wait for activation signals. **As CAF:** Produce ECM. Proliferate. Secrete TGF-β (positive feedback). Contract surrounding matrix.

Core Behaviors

Behavior	Module	PSC	CAF
Signal sensing	M3	Samples TGF-β and SHH	Already activated
Activation transition	M3	Combined signal > threshold → CAF	N/A (irreversible in v1.0)
ECM production	M6	None	High (HA + collagen). Boosted if GLI1=ON
TGF-β feedback	M2	None	Moderate TGF-β secretion (positive feedback loop)
Proliferation	M4	Minimal	Moderate. Boosted if GLI1=ON
Drug response	M7	Drug-indifferent	Drug-indifferent. CAFs persist through treatment.

State Variables

Variable	Type	Initial (PSC)	Mutable?
acta2_active	Boolean	OFF	Yes — once (irreversible)
gli1_active	Boolean	OFF	Yes — reversible (tracks SHH)

Variable	Type	Initial (PSC)	Mutable?
activation_mode	Enum {PSC, CAF}	PSC	Yes — once
ecm_production_rate	Categorical {NONE, HIGH, BOOSTED}	NONE	Yes
tgfb_secretion_active	Boolean	OFF	Yes

The Activation Transition

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PSC (quiescent)          →          CAF (activated)
ACTA2 = OFF                         ACTA2 = ON
ECM production = 0       (one-way)  ECM production = HIGH
TGF-β secretion = 0                 TGF-β secretion = ON
Proliferation = minimal             Proliferation = moderate

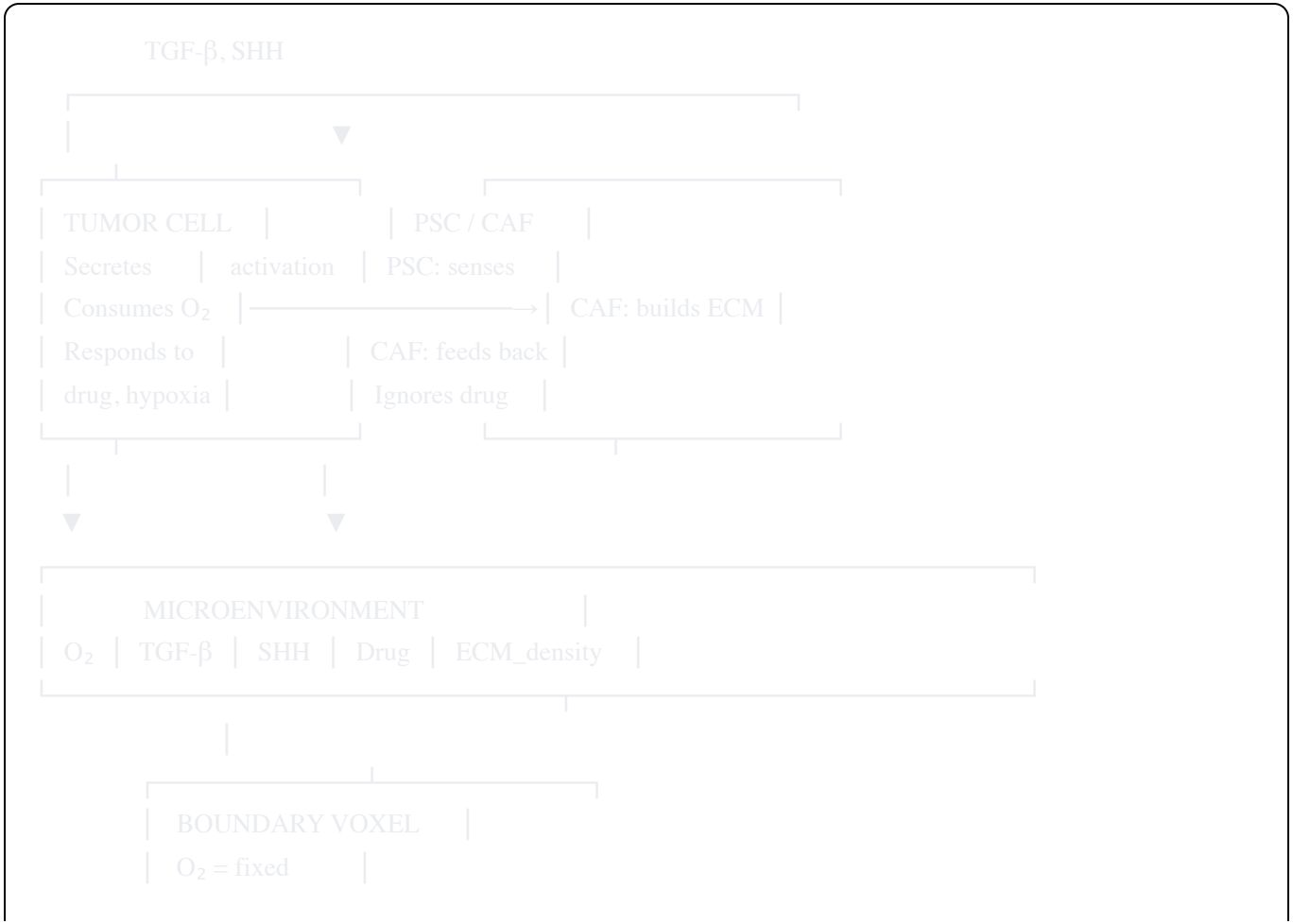
Trigger: local TGF-β + SHH > threshold
Reversibility: NONE in v1.0
```

Actor 3: Boundary Voxel (Inert Source)

Not a cell. A fixed boundary condition representing the vascular supply. Exists only at the domain perimeter.

Behavior	Description
O_2 supply	Fixed Dirichlet boundary value
Drug delivery	Set by EA/scenario; 0 at baseline
Signal sink	TGF-β, SHH absorbed at boundary (Dirichlet ≈ 0)
ECM = 0	No stroma at domain edge

Actor Interaction Summary



Drug = EA-controlled

Who Does NOT Talk to Whom (Critical for correctness)

Non-interaction	Why
Drug → CAF	Standard PDAC chemo does not target CAFs. CAFs persist. Barrier rebuilds.
Immune cells → Anyone	No immune actors in v1.0.
Tumor ↔ Tumor (direct)	All communication via microenvironment fields.

Actor Budget

Peak agents: ~3,000–7,000 cells. Well within PhysiCell's single-node capacity. PDE solving for 5 fields is the computational bottleneck, not agent count.

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§2: WORLD STATE VARIABLES

> 42 values per cell. 9 values per voxel. 18 global scalars. Every one is traceable to a module that reads it, a Reality Anchor that tests it, or both.

Tier 1: Cell-Level Variables (Per Agent)

1.1 Identity & Lineage

`cell_id` (unique), `cell_type` (TUMOR/PSC/CAF), `position` (x,y,z), `parent_id` (for lineage tracing).

1.2 Genotype (Fixed)

gene_state[21] — the 21-gene panel. Each gene in {WT, GOF, LOF}. Fixed at cell birth, inherited by daughters. The root input to the Boolean network.

1.3 Boolean Activity States (Dynamic)

hif1a_active, zeb1_active, cdh1_expressed, acta2_active, gli1_active, nrf2_active, abcb1_active, mmp2_active — updated every time step by module logic. These are "is the gene doing something right now?" as distinct from "is the gene intact?" in §1.2.

1.4 Phenotype Parameters (Derived each step)

proliferation_rate, apoptosis_rate, motility_speed, adhesion_strength, secretion_rates {tgfb, shh}, ecm_production_rates {ha, collagen} — computed from gene states + Boolean states + combination rules + knob values.

1.5 Internal Clocks

intracellular_drug (0–1), time_since_drug_exposure (hours), time_alive (hours, bookkeeping).

Tier 2: Local Environment Variables (Per Voxel)

2.1 Diffusible Substrates (4 PDE fields)

Field	Boundary	Diffuses	Sources	Sinks
oxygen	Dirichlet (fixed)	Yes	Boundary	All cells (consumption)
tgfb	Dirichlet (0, sink)	Yes	Tumor cells, CAFs	Boundary, cellular uptake
shh	Dirichlet (0, sink)	Yes	Tumor cells only	Boundary
drug	Dirichlet (EA-set)	Yes	Boundary (when applied)	Tumor cells (uptake), decay

2.2 Structural Field (1 non-diffusible field)

Field	Diffuses	Sources	Sinks
ecm_density (0–1)	**No**	CAFs (HAS2+COL1A1), mechanical compaction	MMP2 cells, degradation interventions

Plus per-voxel attributes: ecm_ha_fraction (0–1) and ecm_collagen_fraction (= 1 − ha_fraction).

2.3 Derived Local Properties

local_diffusion_coefficient (from ecm_density + ha_fraction) and local_mechanical_stiffness (from ecm_density + collagen_fraction).

Tier 3: Host-Level Variables (Global Scalars)

Population Metrics

total_tumor_cells, total_psc_count, total_caf_count, tumor_caf_ratio

Barrier Metrics

mean_peritumoral_ecm, max_ecm_density, barrier_thickness

Drug & Treatment Metrics

drug_at_tumor_centroid, drug_penetration_half_depth, fraction_tumor_cells_drug_exposed, mean_intracellular_drug, fraction_abcb1_active

Spatial Structure Metrics

tumor_radius, emt_fraction_core, emt_fraction_periphery, o2_at_centroid, survivor_mean_ecm

Simulation Control

`sim_time`, `random_seed`, `intervention_state` (struct tracking active interventions)

Variable Budget Summary

Tier	Count	Bottleneck
Cell-level (per agent)	~42 values × 7,000 cells	Negligible
Environment (per voxel)	~9 values × 250,000 voxels	PDE solver (4 diffusible substrates)
Host-level	~18 scalars	Negligible

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§3: RULES OF THE WORLD

> 69 rules. 55 active in v1.0. 14 deferred but architecturally planned. Written in English, no equations. If the simulation does something not described here, it's a bug.

Part A: Tumor Cell Rules

A1. Proliferation (Rules 1–5)

Rule 1. Tumor cell division rate is determined by genotype via combination rules and calibration knobs. Not negotiated or chosen by the cell.

Rule 2. Severe mechanical compression halts division regardless of genotype. Physical constraint overrides genetic drive.

Rule 3. EMT cells (ZEB1=ON) divide more slowly. Go-versus-grow tradeoff. A penalty, not a block.

Rule 4. Daughter cells inherit full genotype but all Boolean states reset to OFF. Daughters must sense their own environment and activate from scratch.

Rule 5. Daughter cells have zero intracellular drug and a fresh exposure clock. Efflux resistance is not inherited.

A2. Death (Rules 6–8)

Rule 6. Apoptosis probability is low at baseline, increases with intracellular drug concentration.

Rule 7. Dead cells are removed. They stop consuming, secreting, and exerting pressure. No necrotic debris in v1.0.

Rule 8. First drug exposure is most dangerous. Each surviving hour brings the cell closer to efflux activation. Early treatment cycles kill more than later ones.

A3. Mutation & Adaptation (Rules 9–12)

Rule 9. Gene states do not change during a run. No clonal evolution in v1.0.

Rule 10. Phenotypic adaptation is signal-driven and reversible. Cell in hypoxia adapts; cell moved to normoxia reverts.

Rule 11. Drug-induced efflux has persistent memory. ABCB1 doesn't instantly deactivate when drug is removed. NRF2 decays slowly.

Rule 12. [DEFERRED — Cycle 3] Stochastic mutation at division creates intra-tumor heterogeneity.

A4. Secretion (Rules 13–15)

Rule 13. Every living tumor cell secretes TGF-β and SHH constitutively. Cannot be stopped. Consequence of KRAS=GOF.

Rule 14. Hypoxia (HIF1A=ON) amplifies TGF-β secretion. SHH is not hypoxia-responsive. Deeper in the barrier = louder stroma signal.

Rule 15. Secreted signals diffuse outward, slowed by ECM density. Signals concentrate at the tumor-stroma interface.

A5. EMT & Motility (Rules 16–19)

Rule 16. EMT triggers when combined TGF-β (SMAD-independent arm) + HIF1A signal exceeds threshold.

Rule 17. EMT is reversible. Cell moves to low-signal zone → reverts to epithelial. Spatial restriction is maintained by environment.

Rule 18. Mesenchymal cells: fast, loose, MMP2-active, PD-L1 up. Epithelial cells: slow, adherent, no MMP2.

Rule 19. MMP2 degrades ECM locally only. Carves a path at the invasive front, not global barrier dissolution.

A6. The SMAD4 Asymmetry (Rules 20–21)

Rule 20. THE SINGLE MOST IMPORTANT RULE. TGF-β has two signaling arms:

- **SMAD-dependent (brake):** Requires SMAD4. Growth arrest + apoptosis. SEVERED in SMAD4=LOF PDAC.
- **SMAD-independent (invasion):** Does NOT require SMAD4. EMT + motility. INTACT in all tumor cells.

Result: TGF-β builds the wall AND makes the tumor more invasive, but cannot slow the tumor. The brake is gone. The gas pedal works.

Rule 21. Stromal cells always have SMAD4=WT. TGF-β fully activates them. The asymmetry is exclusively a tumor cell property.

Part B: Stromal Cell Rules

B1. Activation (Rules 22–24)

Rule 22. PSC → CAF when combined TGF-β + SHH exceeds threshold. Either alone can trigger.

Rule 23. Activation is irreversible in v1.0. Once ACTA2=ON, it stays ON.

Rule 24. [DEFERRED — Cycle 2] CAF subtypes (myCAF/iCAF/apCAF) with signal-dependent switching.

B2. Barrier Construction (Rules 25–29)

Rule 25. CAFs produce HA (HAS2) and collagen (COL1A1) at their position every time step.

Rule 26. GLI1=ON (SHH pathway active) boosts ECM production. GLI1=OFF (SHH blocked) drops to baseline rate but doesn't stop production.

Rule 27. CAFs secrete TGF-β = positive feedback loop. Stroma recruits itself.

Rule 28. ECM does not diffuse. Stays where deposited. Barrier structure is determined by CAF positions.

Rule 29. ECM turnover is extremely slow. Un-degraded ECM is nearly permanent at simulation timescales.

B3. CAF Persistence (Rules 30–32)

Rule 30. CAFs divide moderately (slower than tumor). Boosted by GLI1.

Rule 31. CAFs do not respond to drug. Drug-indifferent. This is why the barrier rebuilds after treatment.

Rule 32. CAF death rate is effectively zero in v1.0. Activated CAFs persist for the simulation duration.

Part C: Barrier Physics Rules (Rules 33–42)

Rule 33. ECM reduces diffusion of all substrates. More ECM = slower diffusion. Monotonic.

Rule 34. HA impedes diffusion more than collagen (gel resistance, interstitial pressure).

Rule 35. Collagen provides mechanical stiffness more than HA.

Rule 36. ECM traps paracrine signals locally, concentrating TGF-β/SHH at the tumor-stroma interface.

Rule 37. Growing tumor pushes against stiff ECM. Solid stress slows/halts boundary cell division.

Rule 38. Proliferative pressure passively compresses adjacent ECM, increasing density without new production.

Rule 39. Stroma removal releases confinement. Tumor expands into freed space. Without drug, this worsens outcomes.

Rules 40–42. [DEFERRED — Cycle 2] ECM impedes immune cell migration. Collagen alignment guides or blocks infiltration. Barrier buys tumor time to adapt.

Part D: Immune Recognition Rules (Rules 43–48) [ALL DEFERRED]

Rules 43–45. Recognition uncertainty, EMT-driven immune evasion, hypoxia-driven immunosuppression. Architecture pre-planned via IET bookkeeping.

Rules 46–48. Collateral damage from immune effector activity, immune-mediated stroma disruption, host toxicity from excessive immune activation.

Part E: Suppression Rules (Rules 49–54)

Rule 49. Dense ECM physically excludes drug (active v1.0) and immune cells (deferred).

Rules 50, 52–53. [DEFERRED] CAF-mediated immune suppression, PD-L1 checkpoint, immunosuppressive cell recruitment.

Rule 51. Drug efflux (ABCB1) is the second line of defense after the physical barrier. [ACTIVE v1.0]

Rule 54. The Layered Defense Model:

- Layer 1: Physical barrier (stroma) — blocks delivery
- Layer 2: Biochemical suppression — hostile environment
- Layer 3: Cell-intrinsic resistance — efflux, anti-apoptosis

- Layer 4: Regrowth from sanctuary — survivors rebuild

An effective intervention must breach all four. [Layers 1, 3, 4 ACTIVE. Layer 2 partial.]

Part F: Host-Level Rules (Rules 55–65)

Rule 55. Host dies when tumor burden exceeds critical mass (total_tumor_cells > ceiling).

Rule 56. Host death ceiling < domain capacity. Set for lethal pancreatic tumor burden.

Rule 57. [DEFERRED] Host death from treatment toxicity. Approximated as fitness penalty in v1.0.

Rule 58. Host death creates time pressure. Slow strategies that would eventually work but take too long = failure.

Rule 59. Drug concentration has a maximum tolerated dose ceiling. EA cannot exceed it.

Rule 60. Drug delivery is spatially uniform at boundary. Selectivity comes from barrier thickness, not targeting.

Rule 61. EA encodes temporal intervention schedules. Drug pulsing, intermittent SHH inhibition, staggered interventions.

Rule 62. Maximum proliferation rate clamped at biological ceiling (~20h doubling time).

Rule 63. Maximum ECM density clamped at 1.0. No infinite compression.

Rule 64. Volume exclusion enforced by PhysiCell mechanics. Cells cannot occupy the same space.

Rule 65. O_2 cannot exceed boundary value anywhere interior. No internal O_2 sources.

Part G: Meta-Rules (Rules 66–69)

Rule 66. All decisions based on environment state at time step start. Reads before writes. No order-dependent artifacts.

Rule 67. Same random seed = same outcome. ≥3 seeds per condition for anchor/check testing.

Rule 68. Conflict resolution hierarchy: Physics beats signaling beats genotype defaults.

Rule 69. Any behavior not described by a rule does not occur. Closed system. Unexpected behavior is emergent (good) or bug (bad). No third option.

Rule Index Summary

Domain	Active	Deferred
Tumor proliferation	5	0
Tumor death	3	0
Mutation & adaptation	3	1
Secretion	3	0
EMT & motility	4	0
TGF-β / SMAD4	2	0
Stromal activation	2	1
Barrier construction	5	0
CAF persistence	3	0
Barrier physics	7	0
Infiltration	0	3
Immune recognition	0	3
Collateral damage	0	3

Domain	Active	Deferred
Suppression	3	4
Host thresholds	4	0
Treatment constraints	3	0
Biological plausibility	4	0
Meta-rules	4	0
TOTAL	**55**	**14**

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§4: REALITY CHECK SUITE

> Five integration tests. All must pass before EA evolution begins. If any fails, do not proceed.

Check 1: The Untreated Natural History

Setup: 50–100 tumor cells + 200–500 quiescent PSCs. No intervention. 4–6 weeks simulated time. 5 replicates.

Must produce: Barrier self-assembles (peri-tumoral ECM > 0.5), spatial ECM gradient (dense near tumor, sparse at edge), central hypoxia develops (O_2 centroid < O_2 periphery), EMT is peripheral not global, tumor-to-stroma ratio is PDAC-like (tumor < 50% by area), no numerical instability.

If it fails: Paracrine loop broken (check E06, E07, E10), diffusion too fast (check signal diffusion coefficients), domain too small (check O_2 gradient), EMT threshold wrong (check Knob 5a).

Check 2: Standard-of-Care Chemotherapy Fails Realistically

Setup: Mature barrier from Check 1. Drug at max dose for 2 weeks. Withdraw. Observe 2 weeks.

Must produce: Drug reduces tumor (partial response) but does NOT eradicate. Barrier persists through treatment (≥80% of pre-treatment ECM). Survivors concentrated behind dense ECM (spatial sanctuary). ABCB1 resistance emerges. Tumor regrows after withdrawal.

If it fails: Drug too effective (check ECM→drug diffusion), drug ineffective (check M7→M4 kill signal), no regrowth (check survivor genotype integrity), barrier collapsed (check Rule 31 — CAFs must be drug-indifferent).

Check 3: The Vismodegib Paradox Emerges

Setup: Three arms from mature barrier: Control (nothing), SHH inhibition only (no drug), SHH inhibition + drug.

Must produce rank order: Combination (C) < Control (A) < SHH-only (B) for tumor cell count. SHH inhibition alone is worse than doing nothing. Combination is better than either alone.

If it fails: Mechanical confinement too weak (check Module 8, E26), SHH pathway disconnected from ECM (check E07, E11), drug doesn't benefit from thinner stroma (check ECM→drug diffusion coupling).

Check 4: The EA Can Distinguish Good From Bad Strategies

Setup: Four strategies against mature barrier: Drug only, SHH inhibition only, ECM degradation only, ECM degradation + Drug. Compute fitness for each.

Must produce ranking: ECM degrade+Drug (best) > Drug only > ECM degrade only > SHH inhibition only (worst). Consistent across 5 replicates.

If it fails: Fitness function doesn't reward drug penetration improvement (fix fitness formulation), barrier removal looks too good (ECM not rebuilding — check Rule 29, E10).

Check 5: The Sanctuary-Regrowth Cycle Completes

Setup: Mature barrier → best treatment (ECM degrade + drug) for 2 weeks → withdraw → observe 4 weeks.

Must produce five phases: Treatment response (tumor shrinks), residual survival (nonzero floor), early regrowth (cell count climbs), barrier reconstruction (ECM density rises), architectural reconstitution (hypoxia, spatial EMT, stroma ring all return). Drug-hardened cells (ABCB1+) persist in regrowing population.

If it fails: Treatment too effective (reduce degradation intensity, check resistance knobs), no regrowth (check survivor secretion capacity), no barrier rebuild (check CAF persistence — Rule 31, E10, E12).

Suite Execution Protocol

```
Check 1 → PASS → Check 2 → PASS → Check 3 → PASS → Check 4 → PASS → Check 5 → PASS
                                        ↓
                                PHASE 1 VALIDATED ✅
                                EA evolution may begin
                                Cycle 2 scoping may begin
```

Total minimum runs: **50** (5+5+15+20+5). If any check fails >3 times after debugging, escalate to full module wiring review.

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§5: ASSUMPTIONS & EXCLUSIONS

Assumptions (Simplified Biology)

ID	Assumption	Risk	Revisit
A1	All tumor cells share one genotype (no clonal evolution)	MED	Cycle 3
A2	CAF activation is irreversible	LOW	Cycle 2
A3	ECM is scalar density + component fractions, not fiber-resolved	LOW	Cycle 2+
A4	Drug PK is single-compartment and generic	LOW-MED	Agent-specific validation
A5	O_2 supplied from boundary (no explicit vasculature)	MED	Cycle 3
A6	EMT is binary (epithelial/mesenchymal), not continuous spectrum	LOW	If Check 1 EMT pattern fails
A7	Only TGF-β and SHH as stromagenic signals	**HIGH**	If EA finds "too good" strategies
A8	Mechanical forces are scalar pressure, not tensor stress	LOW	If expansion pattern is unrealistic
A9	Time is discrete (fixed time step), not event-driven	LOW	If EA discovers sub-step strategies

Exclusions (Omitted Biology)

ID	Exclusion	Why safe for stroma barrier question	Enters
X1	Immune system	Barrier forms via tumor-stroma signaling, not	Cycle 2

ID	Exclusion	Why safe for stroma barrier question	Enters
		immune interaction	
X2	Vasculature / angiogenesis	Drug access limited by ECM, not vessel density (boundary approximation)	Cycle 3
X3	Metabolism / nutrient competition	Hypoxia-driven adaptation captured by HIF1A; full metabolism is secondary	Cycle 3
X4	Metastasis	Stroma barrier is a local physics problem	Not planned
X5	Neural invasion	Nerves don't produce ECM or modify drug diffusion	Not planned
X6	Exosomes	Dominant signaling is via diffusible cytokines, not vesicles	Unlikely
X7	Lymphatics / interstitial fluid pressure	Captured qualitatively by ECM→diffusion coupling	If quantitative PK needed
X8	Epigenetic regulation	Gene state abstraction (WT/GOF/LOF) covers mechanism-agnostic inactivation	If epigenetic therapy enters EA space

Risk Ranking (Top 3)

1. **A7 (only two stromagenic signals) — HIGH.** Stromagenic redundancy could invalidate EA-discovered pathway inhibition strategies. Mitigation: add PDGF as third signal if EA converges on dual-inhibition strategies that seem too effective.
2. **A1 (monoclonal tumor) — MEDIUM.** If resistance depends on pre-existing subclones rather than inducible adaptation (NRF2/ABCB1), v1.0 overestimates intervention efficacy.
3. **X1 (no immune system) — MEDIUM.** By design, but means v1.0 cannot evaluate strategies requiring immune synergy (stroma depletion + checkpoint inhibitor).

§6: VERSION LABEL & CHANGE CONTROL

Version History

Version	Date	Description
1.0	**2026-02-26**	**Phase 1 complete. Specification locked. All deliverables finalized.**
0.9	2026-02-26	All sections drafted; integration review
0.1–0.8	2026-02-26	Iterative development of individual components

Document Inventory (16 files, ~330KB total)

Category	Files	Total lines
Integrated Spec (this document)	1	This file
Gene-level docs (Gene Set, Gene States, Mutation Map)	3	~950
Cell-level docs (Baseline Traits, Calibration Knobs, Actors)	3	~750
System-level docs (Modules, Edge List, State Variables)	3	~1,100
Validation docs (Reality Anchors, Anchor Metrics, Reality Checks)	3	~900
Policy docs (Pathogenicity Guardrails, Assumptions)	2	~550
Index (TOC)	1	~100

Amendment Protocol

1. **No changes to v1.0 rules, actors, or state variables** without re-running all 5 Reality Checks.
2. **Knob values** (Calibration Knobs doc) may be adjusted during calibration without version increment — these are tuning, not structural changes.
3. **New Reality Anchors** may be added (appended, not modifying existing) without full re-validation; only the new anchor needs testing.
4. **Structural changes** (new modules, new edges, new actors, new rules) require version increment to v1.1 and full Reality Check re-run.
5. **Cycle 2 additions** (immune actors, deferred rules) trigger version increment to v2.0.

Phase 1 Completion Checklist

Benchmark	Deliverable	Status
Mutation→Behavior Class Map (gene panel v1.0)	Stroma_World_Mutation_Behavior_Map_v1.0.md	✅
Tumor baseline knobs + ranges packet	Stroma_World_Tumor_Calibration_Knobs_v1.0.md	✅
Reality Anchors + metrics + pass/fail criteria	Stroma_World_Reality_Anchors_v1.0.md + Stroma_World_Anchor_Metrics_v1.0.md	✅
Module wiring edge list (confidence-based)	Stroma_World_Module_Edge_List_v1.0.md	✅
Integrated Spec: Actors	This document, §1	✅
Integrated Spec: State Variables	This document, §2	✅

Benchmark	Deliverable	Status
Integrated Spec: Rules	This document, §3	✅
Integrated Spec: Reality Check Suite	This document, §4	✅
Integrated Spec: Assumptions/Exclusions	This document, §5	✅
Integrated Spec: Version Label	This document, §6	✅

ALL BENCHMARKS MET. PHASE 1 IS COMPLETE.

This spec is the contract between the simulation and biology. If it's not in here, it's not in Cycle 1.